CraftzyGifts - Project Synopsis (BCSP-064)

Project Title CraftzyGifts - An Online Gifts and Custom Hamper System

1. Introduction and Objectives Introduction: CraftzyGifts is a web-based multi-vendor marketplace for handmade gifts, curated hampers, and custom gifting. Customers can purchase ready-made hampers or build personalized gift boxes. Sellers manage their own product listings and orders. The admin panel provides approval, moderation, and monitoring.

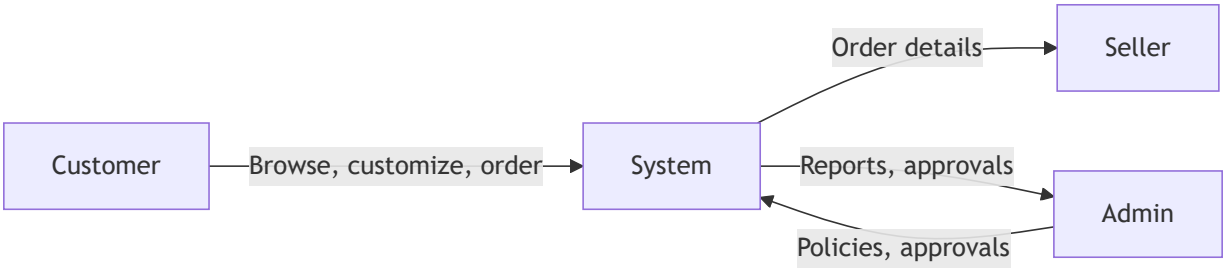
Objectives:

- Allow customers to buy ready-made products or build fully customized hampers.
- Provide home-based crafters an easy platform to sell handmade products.
- Include making and decoration charges for custom hampers.
- Enable reference image upload to guide sellers in designing hampers.
- Provide admin control for monitoring sellers, products, and orders.
- Deliver a responsive, user-friendly web application for seamless browsing and ordering.

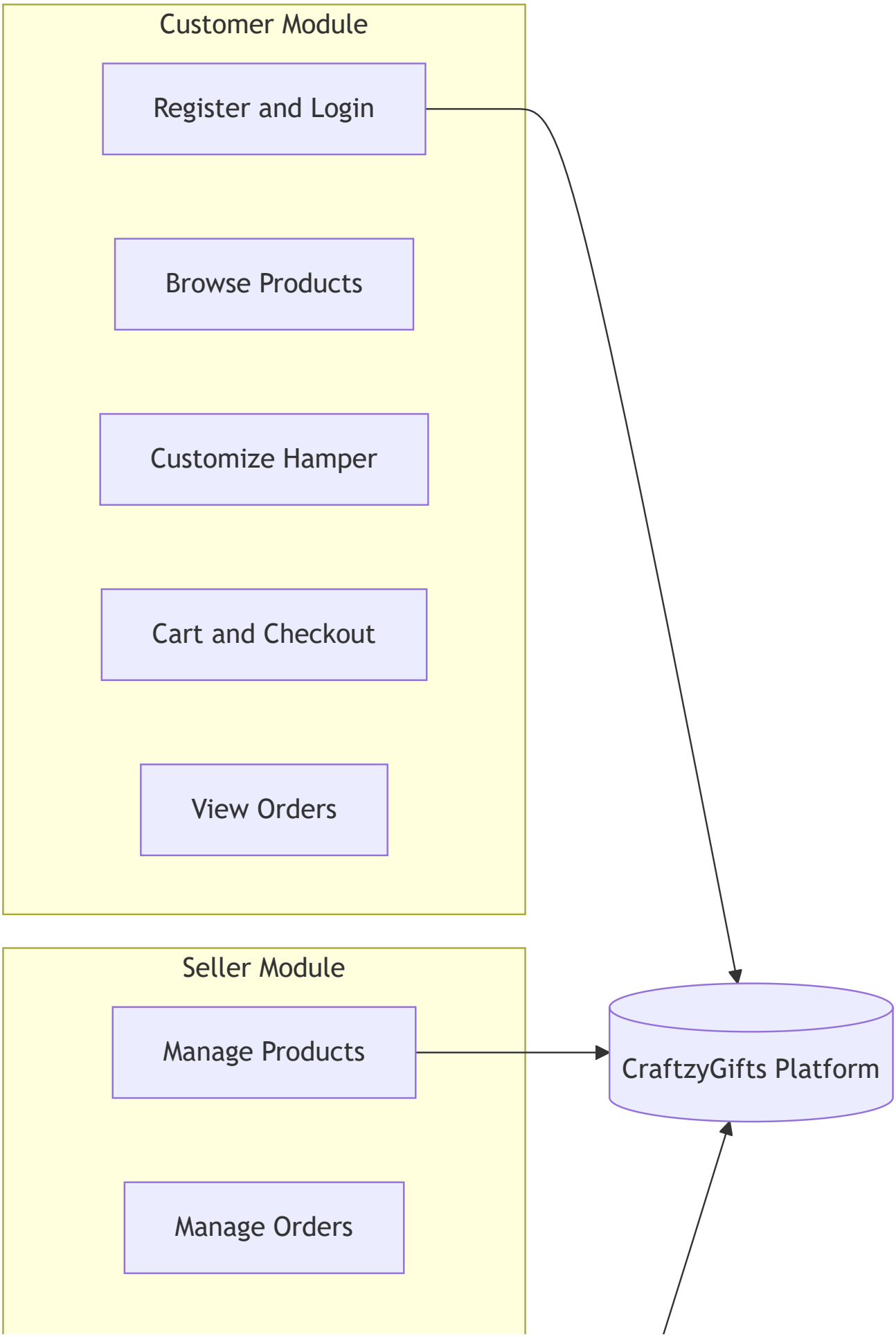
2. Project Category

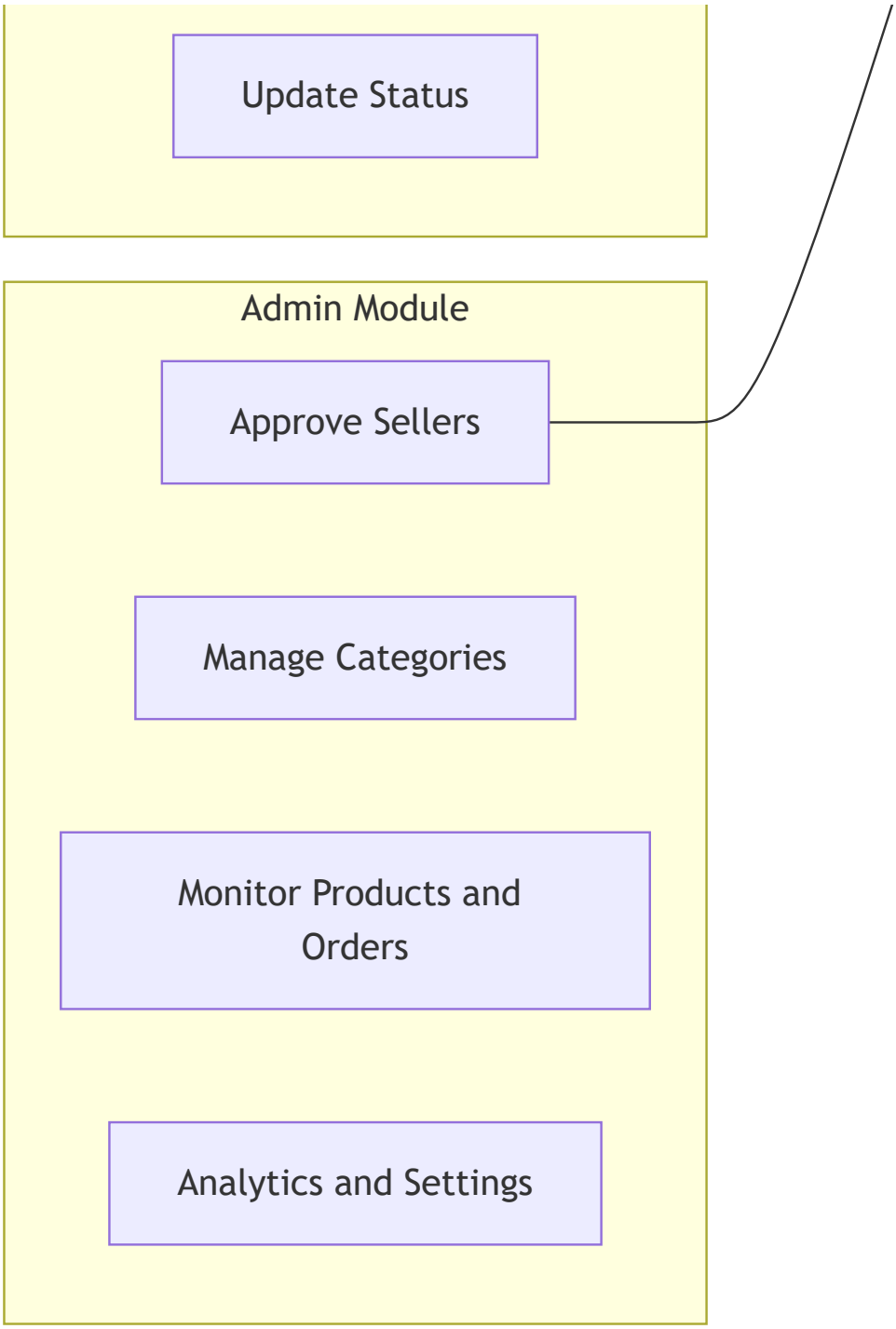
- Category: Web-based multi-vendor e-commerce system with database management.
- Technologies: MERN stack (MongoDB, Express.js, React.js, Node.js).

3. Analysis 3.1 DFD Level 0

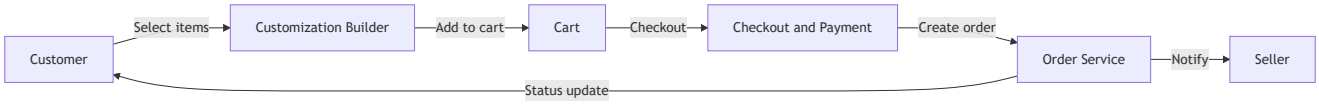


3.2 DFD Level 1

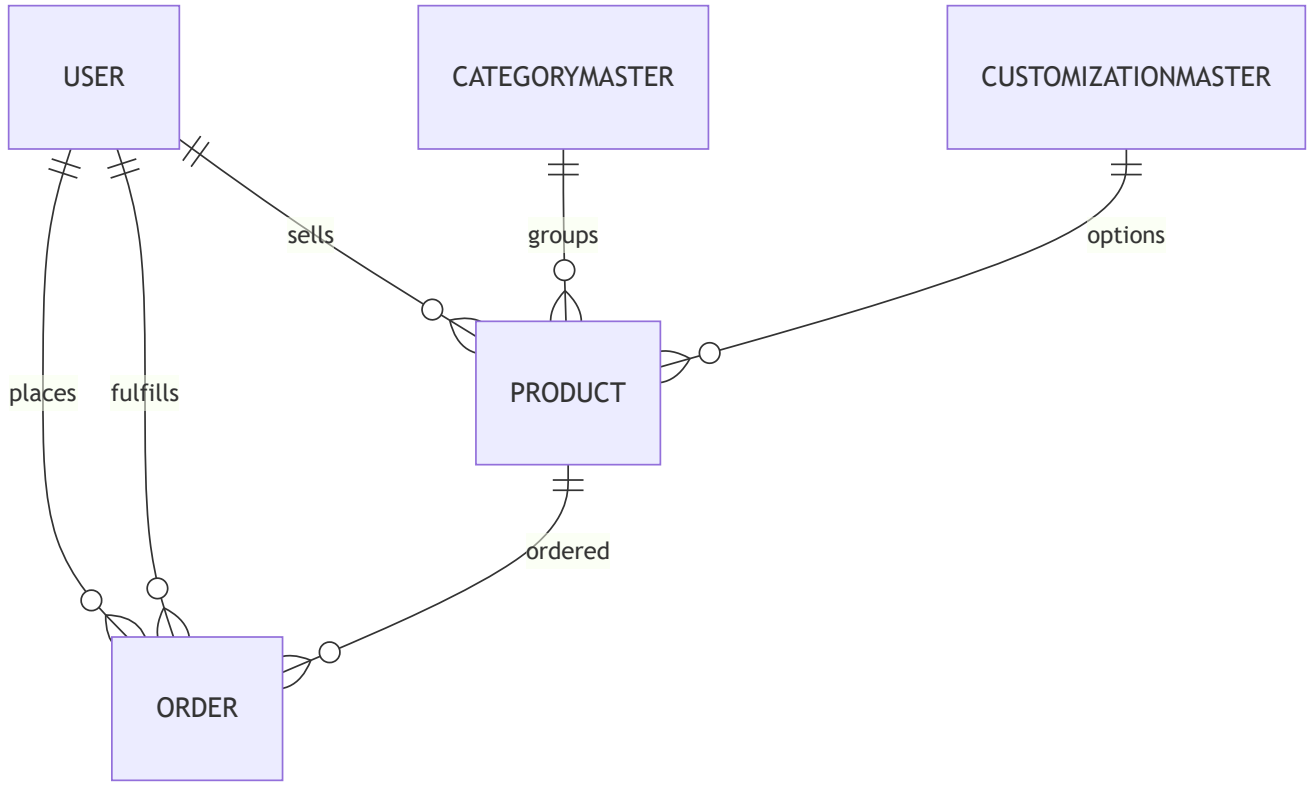

Customer Module
Register and Login
Browse Products
Customize Hamper
Cart and Checkout
View Orders
Seller Module
Manage Products
Manage Orders
CraftzyGifts Platform



3.3 DFD Level 2 (Order and Custom Hamper Flow)



3.4 ER Diagram (High Level)



3.5 Database Design Summary

- User stores customer, seller, and admin accounts with role-based access.
- Product stores listing data, pricing, images, and customization options.
- Order stores purchase, payment, and delivery status.
- CategoryMaster and CustomizationMaster store shared metadata.
- PlatformSettings stores global configuration.

4. Project Structure Modules:

- Authentication module: registration, login, JWT-based access.
- Admin module: seller approvals, category management, platform settings.
- Seller module: product CRUD, order status updates.
- Customer module: browsing, cart, checkout, order tracking.
- Custom hamper module: selection of items, reference images, and making charges.

Data Structures:

- Cart items stored in client-side arrays and persisted in local storage.
- Customization selections stored as lists and key-value maps.
- MongoDB collections for persistent storage.

Process Logic:

- Customer registration and login -> browse or customize -> add to cart -> checkout -> order created.
- Seller updates product inventory and order status.
- Admin reviews sellers and monitors orders and products.

Testing Details:

- Unit testing for modules such as login, add product, cart operations.
- Integration testing for customer -> order -> seller -> admin workflow.

Reports Generation:

- Admin reports for sellers, products, orders, and analytics.
- Seller reports for orders received and earnings.
- Customer order history and customization details.

5. Tools / Platform / Hardware / Software Requirements Software:

- Node.js, npm, MongoDB, React.js, Express.js, VS Code. Hardware:

- Processor: Intel i5 or equivalent.
- RAM: 16 GB.
- Storage: 238 GB or higher. Platform:
- Web-based system, responsive across desktop and mobile.

6. Industry / Client Details This project is developed as an academic project. It can be adapted for small-scale craft businesses or local gift shops.

7. Future Scope and Enhancements

- Mobile apps, international shipping, and multi-language support.
- AI-based recommendations and AR previews for customized hampers.
- Subscription services, advanced seller analytics, and marketing tools.
- AI chatbots, blockchain authenticity, and social integrations.

CraftzyGifts - Project Report (BCSP-064)

Student Name: [Fill] Enrolment No: [Fill] Regional Centre: [Fill] Study Centre: [Fill] Project Guide: [Fill]

1. Introduction CraftzyGifts is a web-based multi-vendor marketplace for handmade gifts and custom hampers. Customers can purchase ready-made products or build personalized hampers. Sellers manage their products and orders, while admins approve sellers, manage categories, and monitor platform activity. The system is designed to demonstrate end-to-end software development following the SDLC.

2. Objectives

- Provide a marketplace for handmade gifts and curated hampers.
- Allow customers to customize hampers with add-ons and reference images.
- Enable sellers to manage listings and order fulfillment.
- Provide admin controls for approval, monitoring, and reporting.
- Deliver a responsive and user-friendly web experience.

3. Tools and Environment Software:

- Node.js, npm
- MongoDB, Mongoose
- Express.js
- React.js, Vite
- Tailwind CSS
- VS Code, Chrome

Hardware (suggested):

- Processor: Intel i5 or equivalent
- RAM: 16 GB
- Storage: 238 GB or higher

4. Analysis Document (SRS) 4.1 Problem Statement Provide an online platform where multiple sellers can list handmade gifts and customers can purchase or customize hampers, with admin oversight for quality and operations.

4.2 Stakeholders and User Classes

- Customer: Browse, customize, buy, and track orders.

- Seller: Create listings, manage inventory, and update order status.
- Admin: Approve sellers, manage categories, and monitor platform health.

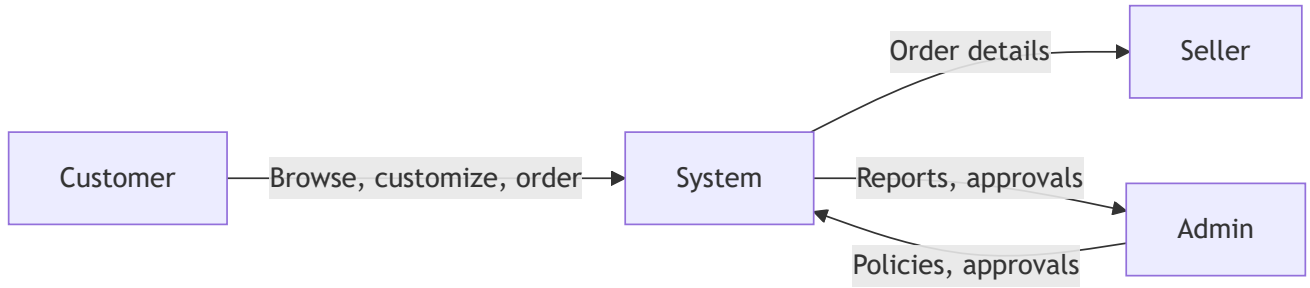
4.3 Functional Requirements

- FR1: User registration and login with role-based access.
- FR2: Customers can browse products by category and seller.
- FR3: Customers can add items to cart and place orders.
- FR4: Customers can create customized hampers with add-ons and reference images.
- FR5: Sellers can add, edit, and remove product listings.
- FR6: Sellers can view and update order status.
- FR7: Admin can approve or reject sellers.
- FR8: Admin can manage categories and monitor products.
- FR9: Admin can view reports for orders, sellers, and analytics.
- FR10: Users can manage profile and addresses.

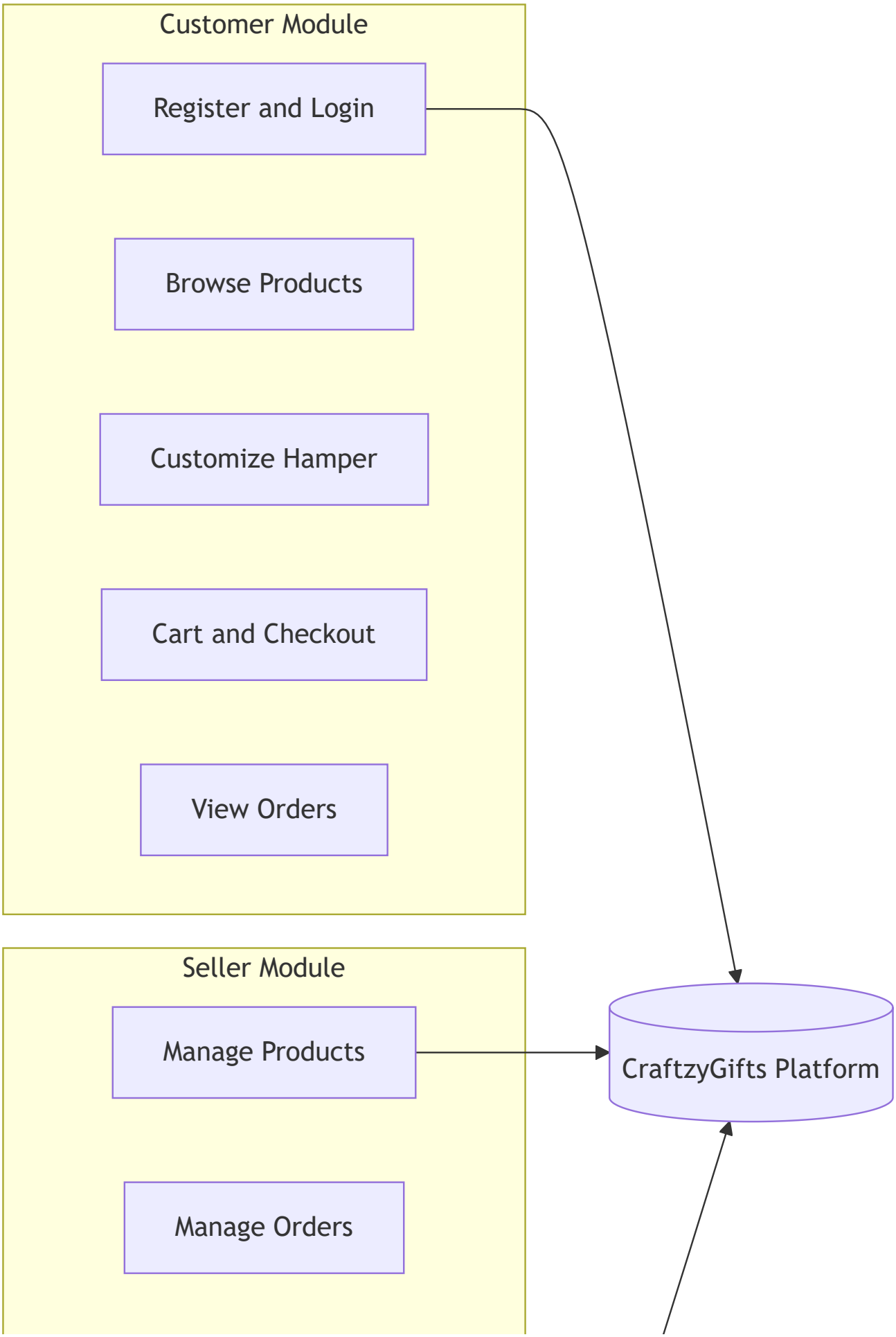
4.4 Non-Functional Requirements

- Security: Password hashing and JWT authentication.
- Performance: Fast browsing and responsive UI for mobile and desktop.
- Reliability: Data consistency using MongoDB and validations.
- Usability: Clear navigation and simple checkout.
- Maintainability: Modular code in client and server layers.

4.5 Data Flow Diagram (Level 0)



4.6 Data Flow Diagram (Level 1)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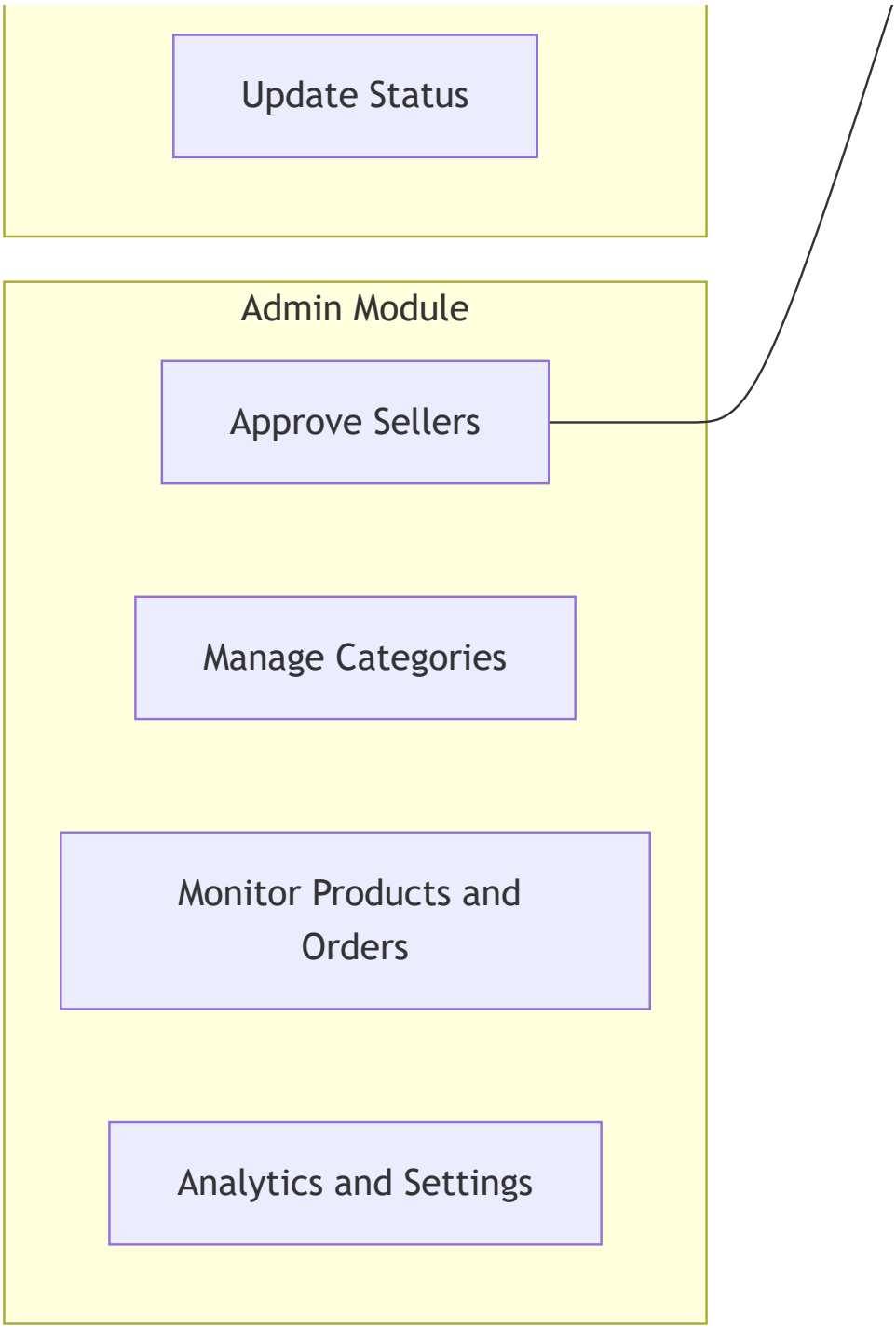
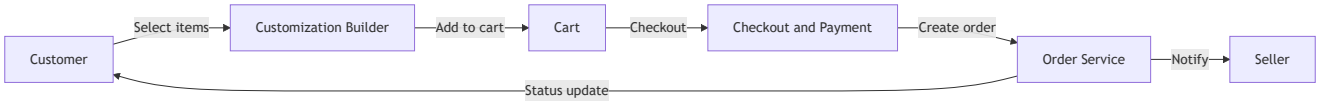
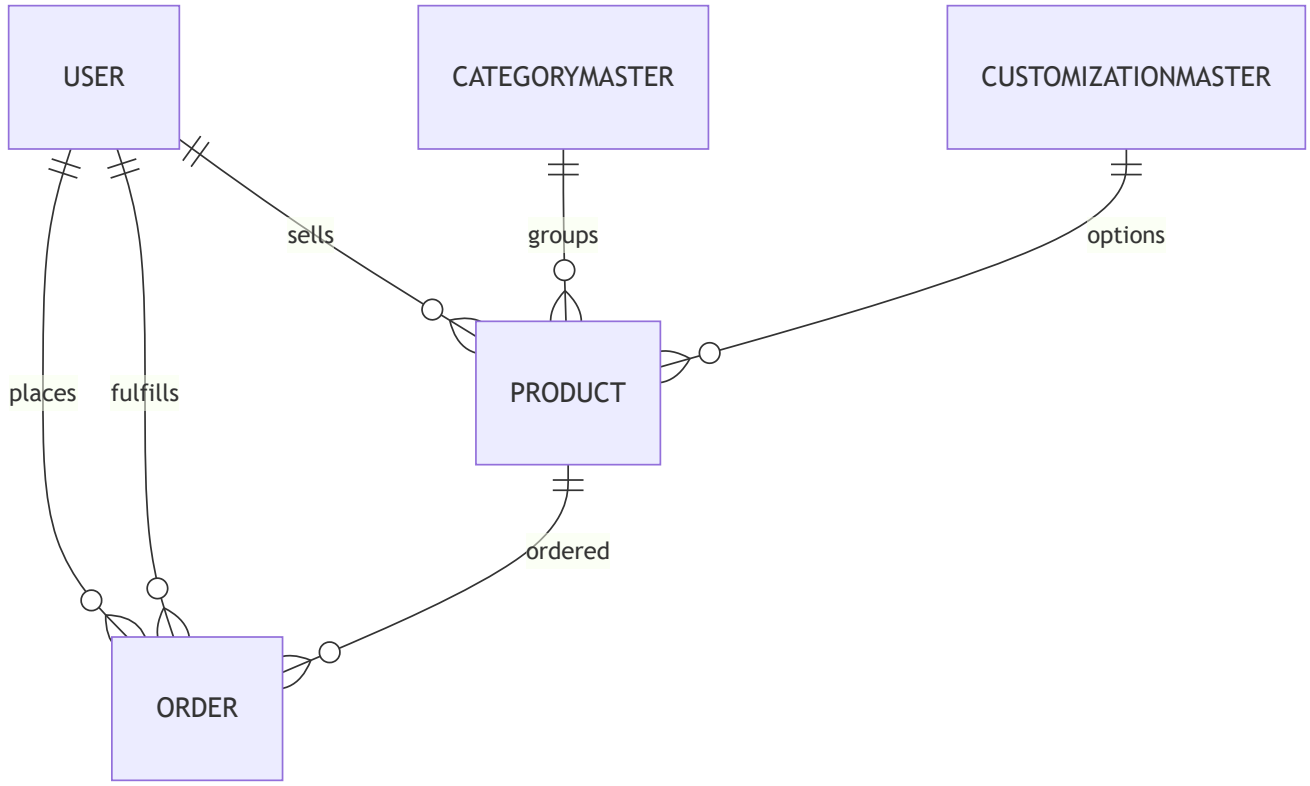


4.7 Data Flow Diagram (Level 2 - Order and Custom Hamper Flow)



4.8 ER Diagram (High Level)



4.9 Data Dictionary (Summary) User

Field	Type	Description
name	String	Full name
email	String	Unique login email
password	String	Hashed password
role	String	customer, seller, admin
sellerStatus	String	pending, approved, rejected
storeName	String	Seller store name
phone	String	Contact number
profileImage	String	Profile image URL
addresses	Object	Shipping and billing addresses

Product

Field	Type	Description
name	String	Product name
description	String	Product details
price	Number	Selling price
stock	Number	Inventory count
category	String	Category label
images	[String]	Image URLs
seller	ObjectId	Reference to User
isCustomizable	Boolean	Customizable or not
customizationCatalog	Array	Customizable items

Field	Type	Description
makingCharge	Number	Additional charge

Order

Field	Type	Description
customer	ObjectId	Buyer
seller	ObjectId	Seller
product	ObjectId	Product reference
quantity	Number	Quantity ordered
price	Number	Unit price
makingCharge	Number	Customization charge
total	Number	Total order amount
status	String	Order state
paymentStatus	String	Payment state
customization	Object	Reference image, notes, selections
shippingAddress	Object	Delivery address

PlatformSettings

Field	Type	Description
platformName	String	Platform title
currencyCode	String	Currency code
lowStockThreshold	Number	Low stock limit
autoApproveSellers	Boolean	Seller auto approval

Additional Collections

- CategoryMaster: category groups and subcategories.
- CustomizationMaster: reusable customization options.
- Notification: seller notifications.
- ContactRequest: customer to seller messages.

5. Design Document 5.1 Modularization

- Authentication module: register, login, JWT validation.
- Customer module: browse, cart, checkout, orders.
- Seller module: product CRUD, order updates.
- Admin module: approvals, categories, analytics, settings.
- Customization module: selections, reference images, pricing.

5.2 Data Integrity and Constraints

- Unique email for users.
- Enumerations for roles, seller status, order status.
- Required fields for core entities (user, product, order).
- Validation for stock and price to be non-negative.

5.3 Procedural Design (Key Flows)

- Login flow: validate credentials -> issue JWT -> set user context.
- Product listing flow: seller submits product -> stored in MongoDB.
- Order flow: cart -> checkout -> order creation -> seller notification.
- Custom hamper flow: select items -> compute making charge -> submit order.

5.4 User Interface Design Customer screens:

- Home, Products, Product Detail, Customization, Cart, Checkout, Orders. Seller screens:
- Dashboard, Products, Listed Items, Orders, Payments, Settings. Admin screens:
- Dashboard, Sellers, Products, Categories, Orders, Reports, Analytics, Settings.

6. Program Code Implementation follows a client-server architecture.

- Frontend: `client/src/` contains pages, components, and utilities.
- Backend: `server/` contains Express routes, controllers, and Mongoose models.
- API base URL: `http://localhost:5000/api`.

7. Testing Test Cases (Manual)

ID	Scenario	Expected Result	Status
TC01	Register new customer	Account created	Pass
TC02	Login with valid credentials	JWT issued	Pass
TC03	Add product to cart	Item appears in cart	Pass
TC04	Place order	Order created with status placed	Pass
TC05	Seller updates order status	Status visible to customer	Pass
TC06	Admin approves seller	Seller status approved	Pass

8. Input and Output Screens Input screens:

- Register, Login, Profile, Manage Addresses, Add Product, Customization. Output screens:
- Product listings, Order history, Seller order list, Admin reports.

9. Security Implementation

- Passwords stored as hashes using bcrypt.
- JWT-based authentication with role-based authorization.
- Request body size limits to prevent oversized uploads.
- Protected routes for seller and admin dashboards.

10. Limitations

- Payment gateway is mock and not integrated with live providers.
- No automated email or SMS notifications.
- Limited analytics and reporting depth for large-scale usage.

11. Future Applications

- Mobile apps and multi-language support.
- AI-based recommendations and AR previews.
- Subscription services and corporate gifting workflows.
- Logistics and payment gateway integrations.

12. Bibliography

- React Documentation

- Node.js Documentation
- Express.js Documentation
- MongoDB Documentation
- Mongoose Documentation
- JSON Web Token (JWT) Documentation

Appendix A - Project Proposal Proforma Attach the approved proforma with signatures and dates.

Appendix B - Project Synopsis Attach the signed synopsis.

Appendix C - Guide Bio-data Attach the guide bio-data with signature and date.

Appendix D - Certificate of Originality (Template) This is to certify that the project report entitled _____ submitted to Indira Gandhi National Open University in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA), is an original work carried out by Mr./Ms. _____, Enrolment No. _____, under the guidance of Mr./Ms. _____.

The matter embodied in this project is a genuine work done by the student and has not been submitted to any other university or institute for the fulfilment of any course of study.

Signature of Student: _____ Date: _____ Signature of Guide: _____ Date: _____